Question			Answer	Marks	AO	Guidance
13	(a)		$\frac{dy}{dx} = p - 16x^{-3}$	M1	3.1a	Uses negative powers to attempt to find expression for $\frac{dy}{dx}$ A term in x^{-3} needed for this mark.
			$p - 16x^{-3} = 0$ $p - 16 \times 2^{-3} = 0$	M1	1.1	Equates their derivative to zero and attempts to solve using $x = 2$
			$p = 2$	A1	1.1	
			When $x = 2, y = 7$ so $7 = 2p + \frac{8}{2^2} + q$ So $2p + q = 5$	M1	3.1a	Uses the given coordinates to link p and q, or their p and q in the Cartesian equation
			so $q = 1$	A1 [5]	1.1	FT their p
13	(b)		$\frac{d^2y}{dx^2} = 48x^{-4} = \left[\frac{48}{x^4}\right]$	B1 [1]	1.1	Allow even from wrong values of p and q
13	(c)		At (2, 7) $\frac{d^2y}{dx^2} = 48 \times 2^{-4} = [3] > 0$	M1	1.1	Substitutes $x = 2$ into their (b) Need not be fully evaluated Also allow for arguing the $48x^{-4}$ is always positive. Do not allow for gradient evaluated on either side of (2, 7)
			So the stationary point is a minimum	A1 [2]	2.2a	Clear statement using the positivity of the second derivative from correct working in (c). FT their second derivative (stating maximum if their value is negative)